

Identification of constitutive arterial tissue parameters using inverse deformation analysis from passive inflation experiments

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Introduction

Changes in the mechanical properties of arteries due to growth, remodeling, or aging are related with cardiovascular diseases. Although arterial constitutive models are promising to quantify these changes, a sufficiently simple model that has some predictive capability has not appeared as of yet [1]. Arterial modeling possesses two great challenges. First, the constitution of arteries are complex mixtures of fluids, fibers and connective tissues. Second, determining the *in-situ* state of stress and the stress-free natural configuration are very challenging tasks. Even in its excised state residual stress remains in the tissue [2, 3]. For any solid constitutive theory to have general applicability, information related with the initial conditions ought to be available. In pressure-myography, the excised arteries are subjected to pre-conditioning by slightly stretching them in order to mimic their *in-situ* state. Although far from being precise, this leads to associating the excised and tethered states (*in-vitro*) with the mostly unknown natural and *in-situ* (initial) configurations of the artery. Parameter estimation can be carried out based on this assumption and the load-displacement data can be obtained from the inflation process. However, the initial stretching of the tethered state needs to be taken into account, but numerous works in literature simply forget or just neglect this step [4, 5, 6].

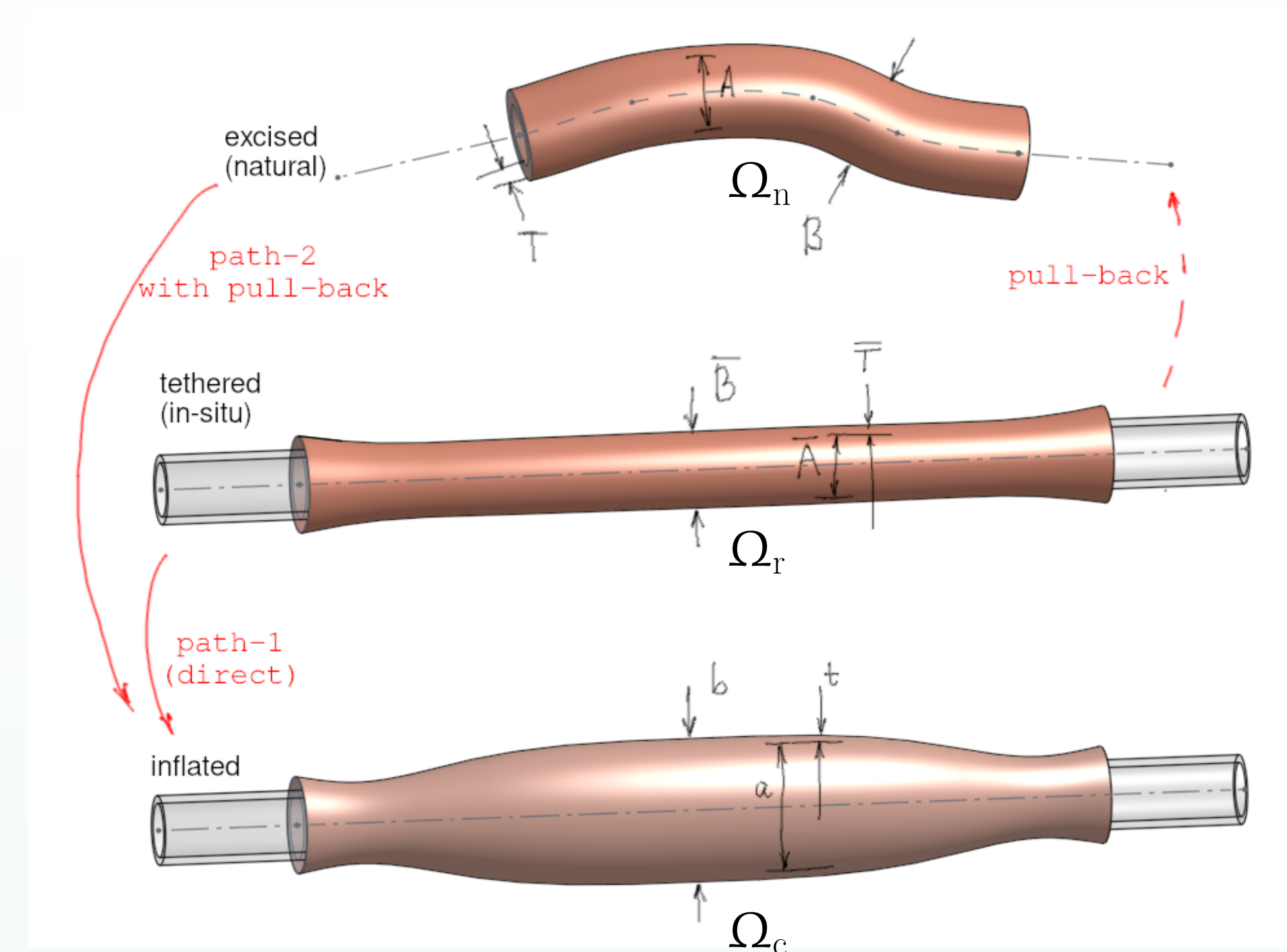


Figure 1: Graphical illustration of the *in-vitro* experimental procedure.

The aim of this study is to illustrate the significance of initial stretching in the tethered state and its effect on the estimated parameters. In order to achieve this, we developed an algorithm that recovers the natural state and compared its results with the case without this algorithm as well as with [7].

In the work of Simon et al (1972), the mechanical response of arterial tissues was studied by using pressure myography based on a sufficiently simple hyperelastic constitutive model proposed by the former authors as well as [8]. The SED function in this model is given as

$$\mathcal{W} = \frac{A}{2k} (e^{k(I-3)} - 1) \quad (1)$$

where A and k are material parameters representing the initial stiffness coefficient and exponential term, respectively, and $I = \lambda^2 + \frac{1}{\lambda}$ is the first invariant of the right Cauchy-Green deformation tensor, and λ is the axial stretch. The initial stress condition in the tethered state was accounted for by introducing a slight modification in the strain energy density (SED) function, by replacing the term $I - 3$ by $I - I_0$, where I_0 is the value of invariant in the tethered (deformed) state. The modified SED function is

$$\mathcal{W} = \frac{A}{2k} (e^{k(I-I_0)} - 1) \quad (2)$$

However, the modified SED (Eq.2) may lead to a difference in the estimated material parameters compared to those that would have been produced using the unmodified SED (Eq.1).

Method

Although there is strong evidence to suggest the opposite, we assume that the excised state to be free of stress, and that in the tethered state (after stretching by $\lambda = 1.53$ as in [7]) the artery is under *in-situ* conditions. In usual experimental practice, the artery is inflated up to a maximum pressure of 200 mmHg (26.6 kPa) as depicted by Path-1 in Figure 1. Most of the recent inverse deformation based parameter estimation studies take the *in-situ* (tethered) state as their reference (initial) configuration [9, 10]. This is especially convenient as the load-displacement data only conveys information regarding the inflation process, while none whatsoever can be found regarding the tethering process. In this study, a new and combined load path was defined as Path-2 (see Fig. 1) to include the stretching process into account. The algorithm of Path-2 including the pull-back step is provided in Figure 2.

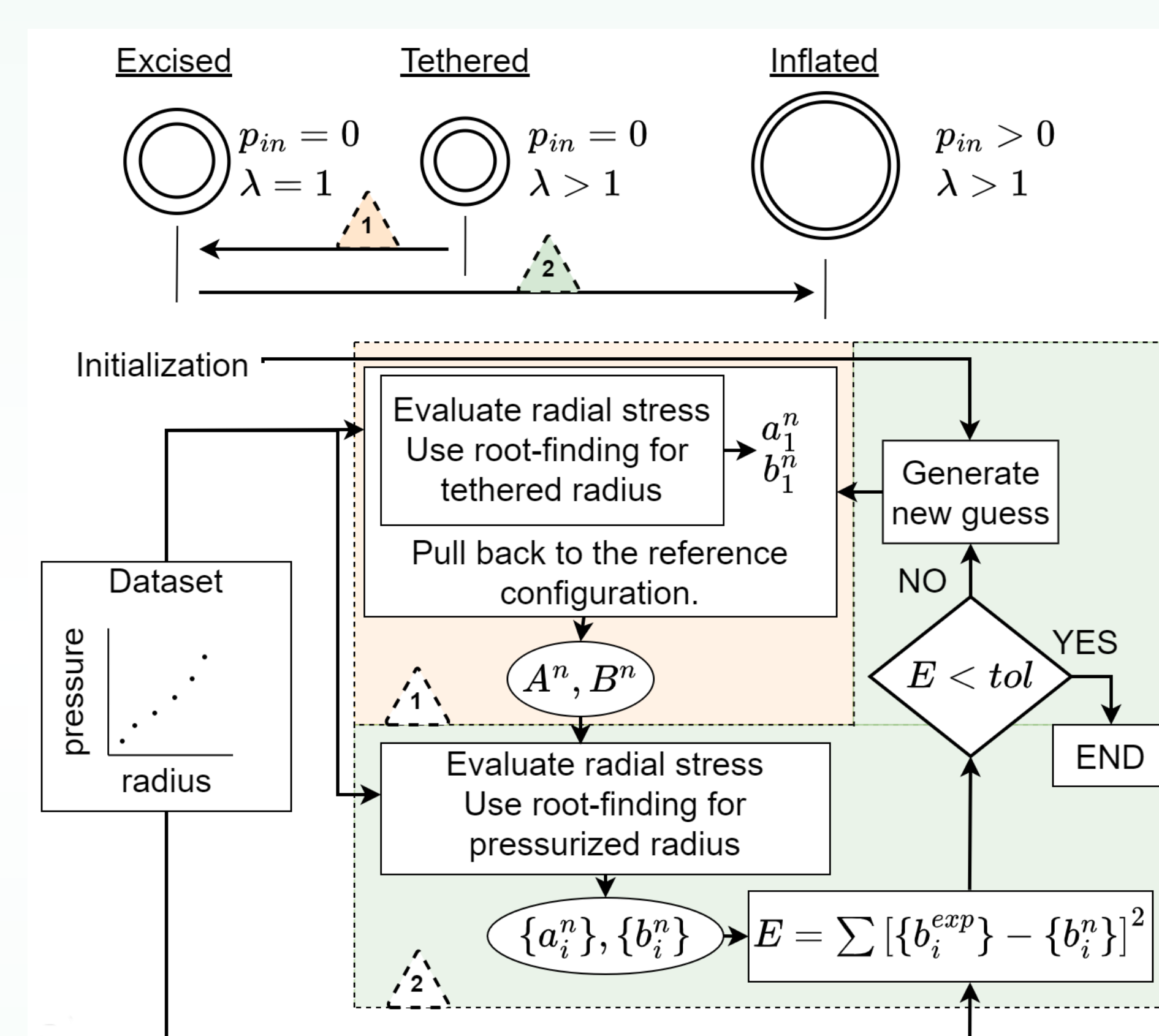


Figure 2: Graphical illustration of the pullback algorithm.

The pull-back is applied to the tethered state as an inverse deformation step in order to obtain an estimate of the excised configuration. Having determined this, the excised state can then be used as the initial configuration and the loading Path-2 can be followed to determine the inflated configuration. However, the SED functions used in these two load paths differ. While Eq.2 is used in Path-1 as in [7], Eq.1 representing an unloaded SED function is used in Path-2.

Results

The material parameters were estimated using the lsqnonlin function found in the optim [11] package in Octave [12] so that the load-displacement curves fitted onto the experimental data points with considerably high coefficients of determination ($R^2 > 0.99$) as shown in Table 1 and Figure 3. The material parameters extracted from Path-1 and Path-2 are tabulated in Table 1. The original parameters obtained in [7] are also shown in the same table.

Table 1: Material parameter estimates from the simulation pair (utilizing the direct and pull-back paths) as well as the reported values in Simon (1972).

Process	A (kPa)	k	R ²
Simon et al.	9.9	2.1	0.9836
Path-1	12.4	1.78	0.9907
Path-2	4.4	1.71	0.9906

The estimated material parameters A and k obtained by traversing Path-1 agreed well with those obtained by [7]. Path-2, how-

ever, produced an initial stiffness parameter, A , which was less than half of that in Path-1. This indicates that using a modified SED function led to overestimation of the initial stiffness parameter.

It is important to note that the pull-back algorithm recovers the natural (excised) configuration with quite a remarkable degree of accuracy. Although the inflated and tethered radii are available in the load-displacement data, radius of the excised state is not. As shown in Table 2 the outer radius of the artery in excised configuration agrees very well with the value reported in [7].

Table 2: Comparison of measured ([7]) and simulated (Path-2) outer radii. * provided in [7]. ** Computed.

Configuration	Outer radius (mm)	
	Experimental	Computed
Excised	3.90*	3.91**
Tethered	3.16	3.16
Inflated	5.30	5.32

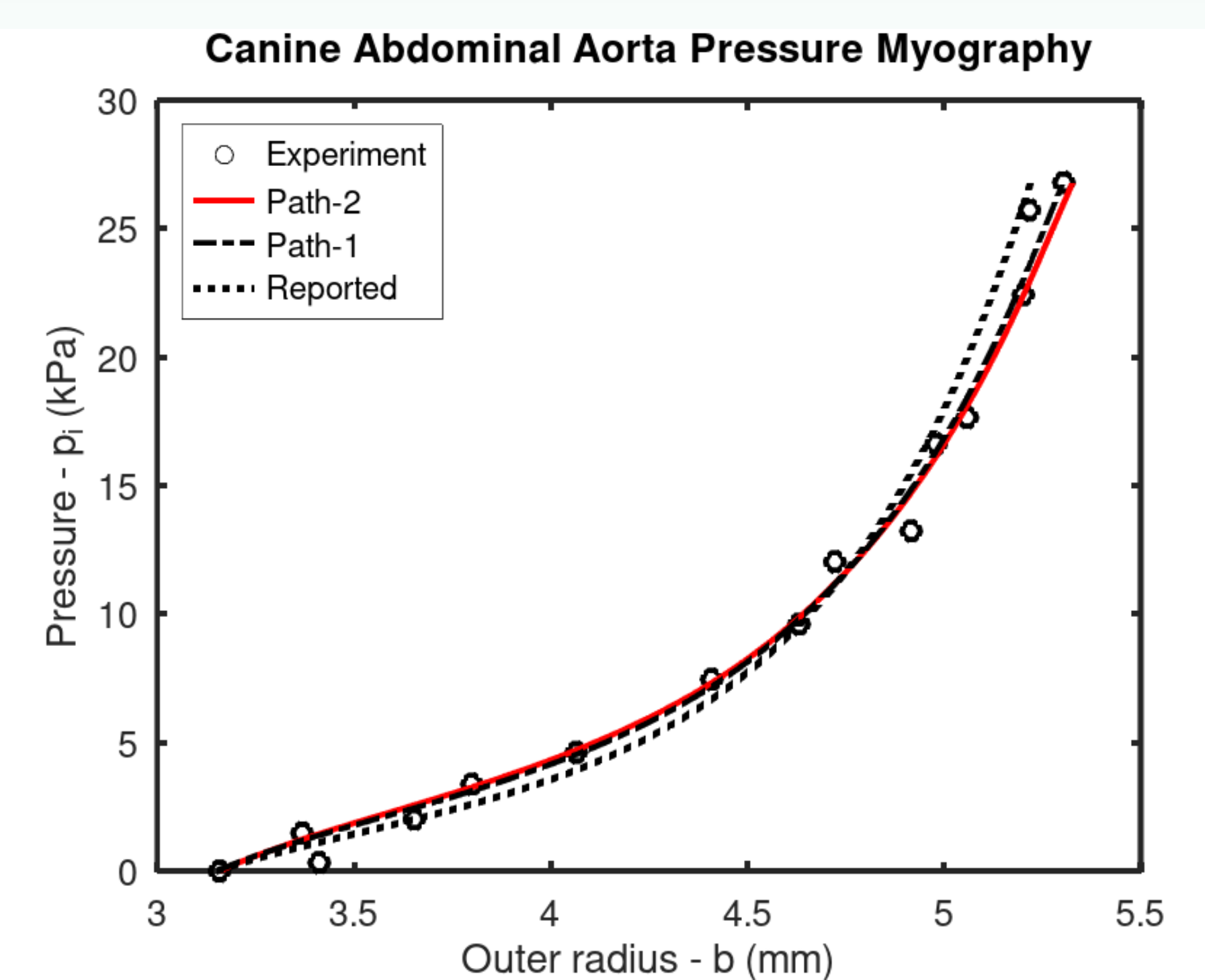


Figure 3: Mechanical response curves obtained using different procedures.

Conclusion

The use of inverse methods for constitutive material parameter estimation is a current topic in arterial mechanics. The inverse methods are not only useful for parameter estimation but they can also be utilized to recover the natural state of an artery. Modifying the SED function to account for the initial deformation in the reference configuration leads to overestimation of the stiffness parameter.

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